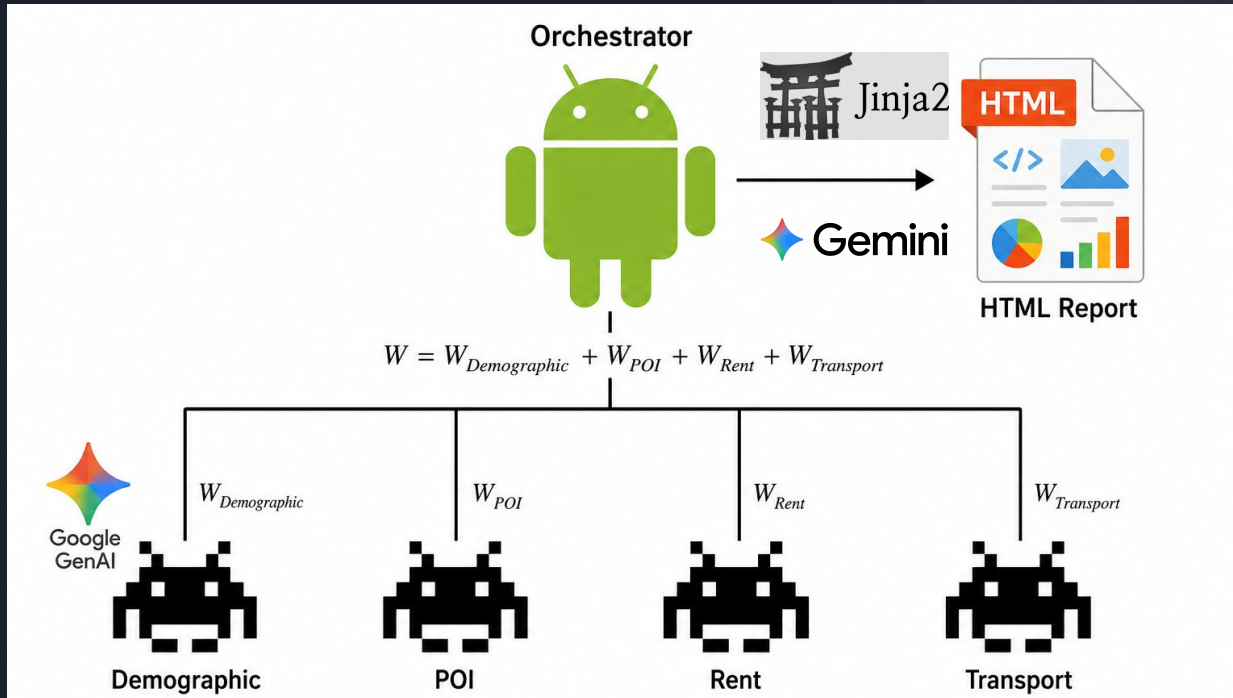


The background is a dark navy blue. In the top-left corner, there are two overlapping geometric shapes: a blue parallelogram and a light green parallelogram. In the bottom-left corner, there is a circular inset showing a detailed, grayscale image of a printed circuit board (PCB) with various electronic components. In the top-right corner, there is a faint, grayscale image of a complex circuit board layout with many traces.

Standort Agent

Architecture Layout





Demographic Agent

Data Scale: Bounded percentage (typically 0.0 to 1.0).

Concept: Translates qualitative business strategy (e.g., "Young Professionals") into a localized quantitative population score.

$$S_{demo} = \sum (w_i \cdot P_i)$$

W is the AI-generated weight for a specific age bracket. It interprets the context of the “zielgroup” and give a weight value to the age brackets.

Where i in $\{18_34, 35_54, 55_plus\}$ and $\sum w_i = 1.0$



Point of Interest Agent (POI)

Data Scale: INDEX (typically 0.0 to 10.0).

Concept: The POI refers to the commercial and infrastructure footprint of the location.

Logic: Pass the POI index as it is. Since is a clear indicator.

$$S_{\text{poi_raw}} = D_{\text{local}}$$

D local is the direct POI index provided in the municipal data.



Rent Agent

Data Scale: Eur/m2 (typically 0.0 to infinite).

Concept: Uses the mietindex_eur_m2 (from dataset), flaeche_m2, budget_miete_eur (from profile)

Logic: Compute the expected rent.

$$E = \text{mietindex_eur_m2} \times \text{flaeche_m2}$$

Then the score is the remaining budget.

$$S_{rent} = \text{budget_miete_eur} - E$$

The score is negative if the expected rent exceeds the budget.



Transport Agent

Data Scale: INDEX (typically 0.0 to 10.0).

Concept: The `oev_score` indicates how good is the public transport, the higher the better.

Logic: We pass the as it is.

$$S_{trans} = \text{oev_score}$$



Orchestrator

Normalization and Weighting

orchestrator collects the raw scores (S) from all four agents for a specific municipality, it must standardize them using Min-Max Normalization. Where x is the scores from the agents.

$$x_{norm} = \frac{x - \min(X)}{\max(X) - \min(X)}$$

X_norm: [0,1]

Then it applies the Weighted Sum to get the Final Ranking Score F.

Where w_i is a fixed weight (0.25) that give a predefined importance to the agents scores.

$$F = \sum_{i=1}^4 (w_i \times x_{norm,i})$$